

Predicting Air Quality Categories in India Using Machine Learning



Capstone Lightning Talk 2 | EDA, Cleaning & Modeling Plan

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Refined Problem Statement

Air pollution in Indian cities poses serious public health risks, but AQI-based warnings are often issued reactively rather than in advance. Using daily pollutant readings collected across Indian cities and monitoring stations, **can we classify a day's air quality into its AQI category (Good, Satisfactory, Moderate, Poor, Very Poor, Severe) to help authorities and citizens take preventive action before exposure becomes dangerous?**

Success Metric

Macro F1-score, to account for **class imbalance** while still valuing performance on high-risk classes (Poor, Very Poor, Severe).

Why Classification, Not Regression?

Considered predicting AQI directly, but AQI_Bucket is the actionable decision layer authorities and citizens respond to — and AQI is itself a derived formula from the pollutants.

Potential Audience

1

Public Health Authorities

Issue preventive advisories before pollution reaches dangerous levels, rather than reacting after the fact.

2

City & Environmental Planners

Identify pollution patterns across locations and seasons to guide policy and monitoring investment.

3

Citizens in High-Pollution Cities

Make daily exposure decisions — outdoor activity, masks, air purifiers — based on forecasted risk category.

Goals & Objectives

Problem Type

Multiclass Classification (6 classes)

Planned Models

- Logistic Regression — interpretable baseline
- Random Forest — robust to skew and imputation noise
- KNN or SVM — comparison point

Pipeline

scikit-learn Pipeline + GridSearchCV with StratifiedKFold

Evaluation Metrics

Macro F1-Score

Primary metric — accounts for class imbalance

Per-Class Recall

Focused on Poor, Very Poor, and Severe — the classes that matter most for public health action

Baseline

Majority-class baseline (always predicting “Moderate”) to contextualize model lift

About the Data

102,115

Rows (post-cleaning)

15

Columns

135

Monitoring Locations

6

AQI Bucket Classes

Data Source

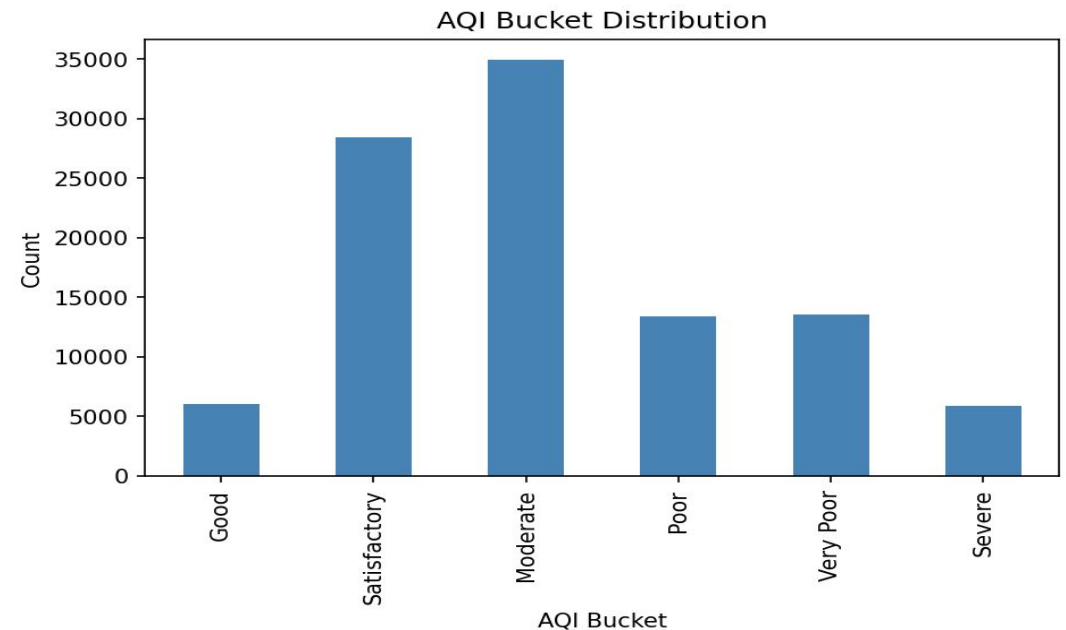
Kaggle: Air Quality Data in India (city_day + station_day, merged and deduplicated)

Target: AQI_Bucket

Good · Satisfactory · Moderate · Poor · Very Poor · Severe (moderately imbalanced)

Key Features

13 pollutant concentrations (PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene, Toluene) + Location + Date



AQI Bucket distribution — target class imbalance

Handling Missing Data

Pollution is seasonal, so gaps were filled based on nearby days.

1

Drop unusable columns

Xylene was ~74% missing — too sparse to impute responsibly, so it was dropped entirely.

2

Interpolate by location × date

Missing pollutant values were filled using linear interpolation within each location, capped at a 5-day gap based on observed spacing.

3

Location median fallback

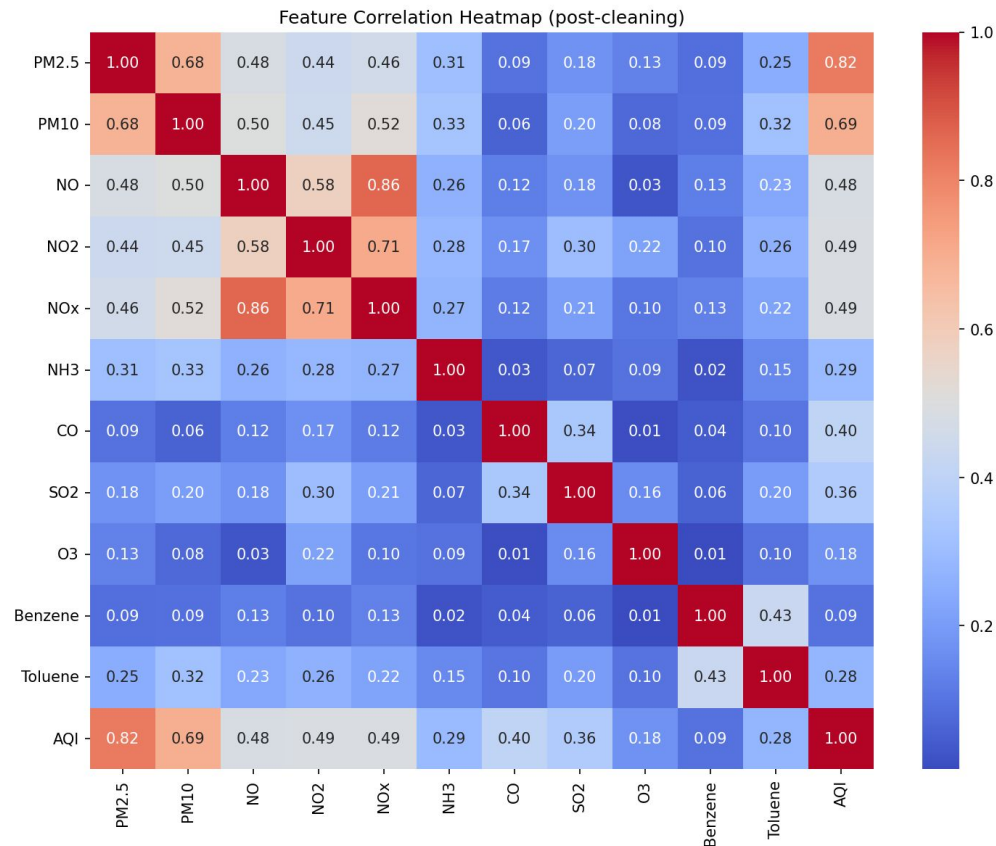
Gaps too large to interpolate (start/end of a series, or long stretches) were filled with that location's own median.

4

Global median fallback

Some stations never measured a pollutant at all (e.g. 18 stations for PM10). These were filled using the dataset-wide median as a last resort.

EDA: Feature Correlations

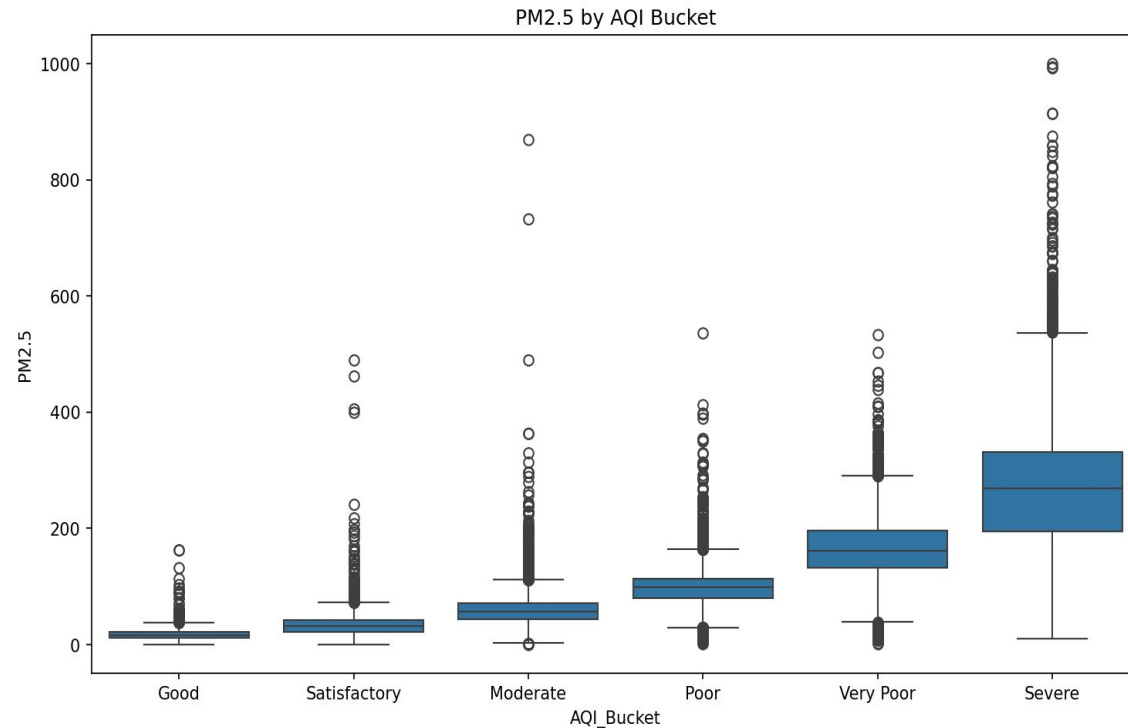


Correlation matrix — continuous pollutant columns

Key Takeaways

- **PM2.5** is the strongest predictor of AQI (0.82)
- PM10 dropped from 0.87 to 0.69 — likely diluted by imputed values
- NO and NOx are highly correlated (0.86), as expected since NOx includes NO
- AQI itself will be dropped before modeling to avoid data leakage

EDA: PM2.5 Separates Classes Cleanly



PM2.5 by AQI Bucket

Key Takeaways

- Median PM2.5 climbs cleanly from ~15 (Good) to ~270 (Severe)
- Outliers were checked individually — they reflect real extreme pollution days (e.g. Delhi in peak winter), not sensor errors, so they were kept rather than removed
- Only ~3.7% missing before imputation, so this signal is trustworthy
- Right-skew means scaling or a log-transform is planned for distance-based models

Progress Report: Challenges

1

Hidden Duplicates

~19% of merged rows were duplicated between city-level and station-level data for single-station cities — identified and removed.

2

Missing Values

Handled via time-based interpolation (5-day gap cap) within each location, with median fallbacks for longer gaps.

3

PM10 Reliability

~23% of PM10 values are global-median fallbacks (18 stations never measured it) — weaker signal than PM2.5, flagged for modeling.

4

Seasonal Pattern Confirmed

Chi-square test on Season vs. AQI_Bucket came back highly significant ($\chi^2 \approx 24,028$, $p < 0.001$) — confirms pollution category shifts by season, supporting Season as a planned feature.

Next Steps

- 1 Finalize categorical encoding strategy for Location
- 2 Engineer Month / Season features from Date
- 3 Add an imputation flag for PM10 (PM10_was_imputed)
- 4 Build preprocessing pipeline + run GridSearchCV across planned models
- 5 Compare model performance against majority-class baseline